

Modeling of Flow Blockage in a Liquid Metal-Cooled Reactor Subassembly with a Subchannel Analysis Code

Hae-Yong Jeong, Kwi-Seok Ha, Won-Pyo Chang, Young-Min Kwon & Yong-Bum Lee

To cite this article: Hae-Yong Jeong, Kwi-Seok Ha, Won-Pyo Chang, Young-Min Kwon & Yong-Bum Lee (2005) Modeling of Flow Blockage in a Liquid Metal-Cooled Reactor Subassembly with a Subchannel Analysis Code, Nuclear Technology, 149:1, 71-87, DOI: [10.13182/NT05-A3580](https://doi.org/10.13182/NT05-A3580)

To link to this article: <https://doi.org/10.13182/NT05-A3580>



Published online: 10 Apr 2017.



Submit your article to this journal [↗](#)



Article views: 168



View related articles [↗](#)



Citing articles: 9 View citing articles [↗](#)

MODELING OF FLOW BLOCKAGE IN A LIQUID METAL-COOLED REACTOR SUBASSEMBLY WITH A SUBCHANNEL ANALYSIS CODE

THERMAL HYDRAULICS

KEYWORDS: flow blockage, sub-channel analysis, liquid metal-cooled reactor

HAE-YONG JEONG,* KWI-SEOK HA, WON-PYO CHANG,
YOUNG-MIN KWON, and YONG-BUM LEE
Korea Atomic Energy Research Institute, P.O. Box 105
Yuseong, Daejeon 305-600, Korea

Received March 22, 2004

Accepted for Publication July 2, 2004

The local blockage in a subassembly of a liquid metal-cooled reactor (LMR) is of importance to the plant safety because of the compact design and the high power density of the core. To analyze the thermal-hydraulic parameters in a subassembly of a liquid metal-cooled reactor with a flow blockage, the Korea Atomic Energy Research Institute has developed the MATRA-LMR-FB code. This code uses the distributed resistance model to describe the sweeping flow formed by the wire wrap around the fuel rods and to model the recirculation flow after a blockage. The hybrid difference scheme is also adopted for the description of the convective terms in the recirculating wake region of low velocity. Some state-of-the-art turbulent mixing models were implemented in the code, and the models suggested by Rehme and by Zhukov

are analyzed and found to be appropriate for the description of the flow blockage in an LMR subassembly. The MATRA-LMR-FB code predicts accurately the experimental data of the Oak Ridge National Laboratory 19-pin bundle with a blockage for both the high-flow and low-flow conditions. The influences of the distributed resistance model, the hybrid difference method, and the turbulent mixing models are evaluated step by step with the experimental data. The appropriateness of the models also has been evaluated through a comparison with the results from the COMMIX code calculation. The flow blockage for the KALIMER design has been analyzed with the MATRA-LMR-FB code and is compared with the SABRE code to guarantee the design safety for the flow blockage.

I. INTRODUCTION

In a liquid metal-cooled reactor (LMR) the fuel pins are configured very compactly, and the heat flux from the fuel rods is usually higher than that of a pressurized water reactor. In spite of the narrow coolant channels, the heat generated in the fuel rods is successfully transported to the heat exchangers because of the high heat capacity and good heat transfer characteristics of the liquid metal coolant. Therefore, it is important to maintain the integrity of the fuel rod configuration for the continuous operation of a liquid metal-cooled reactor. To maintain the cooling geometry and to prevent the fuel rods from contact, a thin wire is usually wrapped helically around each rod along its axis. Another role of the wire wrap is to promote

mixing between the subchannels by generating an active cross flow and a swirl flow at the outer region of an assembly.

Because of the compact design of an LMR core, the local blockage in a subassembly of an LMR is of particular importance. When an obstacle or a blockage is formed in the flow path, the local temperature of the coolant increases downstream of the blockage and the integrity of the fuel clad can be threatened. The flow blockage can be formed either by the introduction of foreign material remaining in the primary coolant system or by the degraded fuel itself. The early detection of the flow blockage is a primary concern because the event could initiate a severe core disruptive accident when any preventive action, such as a fast reactor shutdown, does not follow, automatically or manually. For this reason, it is necessary to predict the temperature distribution within the assembly with a flow blockage using a reliable computer code

*E-mail: hyjeong@kaeri.re.kr

applicable to the situation. The prediction results calculated by the code are utilized for the safe design and operation of an LMR.

Since the blockage disturbs the normal flow field, it is important to simulate the flow and temperature fields correctly in the analysis of a flow blockage. When a blockage occurs in the flow path, the axial flow rate decreases drastically up to a certain distance downstream from the blockage, and a recirculating wake is formed. This recirculating wake is characterized by a reverse flow and a large lateral cross flow, and it is accompanied by an increase of the local temperatures of the coolant and the fuel rod surface. Therefore, the analysis code for the flow blockage requires a proper numerical scheme and thermal-hydraulic model to deal with the recirculating flow.

II. PREVIOUS WORKS

Many countries leading in the development of LMR technology have been performing a lot of experimental and analytical research on flow blockage because of its importance to the core safety of an LMR. The analytical approach can be divided into two large categories, the subchannel approach and the porous medium approach. In a subchannel approach the lateral exchanges between the adjacent subchannels are simplified and the direction of the lateral flow is not necessarily specified. One of the major advantages of the subchannel approach is its convenience in reflecting the thermal-hydraulic and geometrical characteristics of the flow in a rod bundle. The subchannel approach basically assumes that the axial flow is predominant over the transverse flow.

Some subchannel codes such as COBRA-IV (Ref. 1) developed at the early stage of the study were restricted to forward flow and thus were unable to model the reverse flow downstream of the blockage. This restriction mainly arises from the solution procedure that calculates the flow and temperature along the flow direction with a marching stepup axially. To overcome the restriction, more advanced numerical methods with more accurate physical models were adopted in the following subchannel codes, such as SABRE (Ref. 2), ASFRE (Ref. 3), and THI3D (Ref. 4). These codes have been applied to the analysis of a flow blockage successfully.

The porous medium approach provides an effective representation of the continuum flowing in solid objects, for example, large fuel bundles or the shell side of steam generator tubes. The COMMIX series are the best known codes using the porous medium approach.⁵ However, the preparation of the code input is so complicated and it takes so much time to get the solution with a porous medium approach that the application of the porous medium codes to the flow blockage analysis is not generally efficient. Therefore, the subchannel approach is preferably adopted for the core thermal-hydraulic design and some safety analyses of a liquid metal-cooled reactor.

In Korea the conceptual design of the Korea Advanced Liquid Metal Reactor⁶ (KALIMER), which is a sodium-cooled pool-type reactor, has been completed. For the analysis of the flow blockage in the KALIMER core, the subchannel analysis code of MATRA-LMR-FB is being developed by modifying some models in the MATRA-LMR code. The MATRA-LMR code⁷ has been developed based on the COBRA-IV and the MATRA code⁸ for the thermal-hydraulic design of the KALIMER core. However, the MATRA-LMR code is not suitable for analyzing the thermal-hydraulic phenomena associated with a flow blockage because the lateral cross flow is determined from the forced cross-flow model, in which the transverse mass flow rate is given as proportional to the axial subchannel mass flow rate. Further, the upwind spatial differencing for the convective terms in the MATRA-LMR code results in a remarkable numerical diffusion in the region of the low flow velocity, such as the recirculating wakes after a blockage. The other drawback of the MATRA-LMR code is found with the turbulent mixing model, which requires one or more user-selective constants, which introduces the inevitable user dependence. All these limitations of the MATRA-LMR code in the modeling of the flow blockage have been removed in the development of the MATRA-LMR-FB code.

III. FLOW BLOCKAGE MODELING IN THE MATRA-LMR-FB CODE

In the MATRA-LMR-FB code, the influence of the wire wrap is modeled with a distributed resistance model suggested by Ninokata et al.³ instead of the forced cross-flow model. The hybrid difference scheme is implemented to minimize the numerical diffusion in the region of a low velocity. The state-of-the-art turbulent mixing models are also implemented to remove the user dependence in the modeling of the turbulent mixing.

III.A. Distributed Resistance Model

The axial and transverse momentum conservation equations for a subchannel analysis code are written as follows:

$$\begin{aligned} \frac{\partial}{\partial t} \langle \rho u \rangle_V A + \frac{\partial}{\partial x} \langle \rho u^2 \rangle_A A + [D_C^T] \langle \rho u v \rangle_S S \\ = -A \frac{\partial}{\partial x} \langle p \rangle_A - A \langle \rho \rangle_A g - f_T [D_C^T] \{w'\} [D_C] \{u'\} \\ - \frac{1}{2} \left(\frac{f}{D_h} + \frac{K}{\Delta x} \right) \langle \rho u^2 \rangle_A A \end{aligned} \quad (1)$$

and

$$\begin{aligned} & \frac{\partial}{\partial t} \langle \rho v \rangle_{V'} S + \frac{\partial}{\partial x} \langle \rho v u \rangle_{A'} S \\ & + C_S [D_C] [D_C^T] \left\{ \frac{S}{l} \langle \rho v^2 \rangle_S \cos \beta \right\} \\ & = - \frac{S}{l} [D_C] \{ \langle p \rangle_A \} - \frac{1}{2} \frac{S}{l} K_G \langle \rho v^2 \rangle_S, \quad (2) \end{aligned}$$

where β is the angle between a communicating gap and the gap of interest and the other terms are as described in the Nomenclature (p. 85).

The last terms of the right-hand side in Eqs. (1) and (2) represent the momentum exchange between the solid surface and the fluid due to the forces exerted on the fluid by the wall such as the fuel rods and the wire wraps. In traditional subchannel analysis codes such as FULMIX, ORRIBLE, and ASFRE-II and even in some recently developed subchannel codes such as MATRA-LMR, the transverse mass flow rate is not obtained directly by solving the transverse momentum conservation of Eq. (2) because it is a very complicated problem to treat the flow resistance induced from the wire wrap properly. The transverse momentum balance is used only to solve the pressure variable and to estimate the initial conditions for the given time step. Instead of directly solving the conservation equation, the transverse mass flow rate was given as proportional to the axial flow rate, assuming that all the flow is formed in parallel along the path of the wire wrap.

Apparently this forced cross-flow model is based only on continuity considerations and does not account for the momentum effects; thus, it does not fully describe the actual flow behaviors observed in the experiments of the flow in wire-wrapped bundles. To account for the presence of the wire wrap spacer in a rod bundle and to represent the direction and resistance characteristics correctly, Davis⁹ developed the distributed resistance model for the SABRE code, which was applicable for the predominantly axial flows under turbulent flow conditions. Recently, Ninokata et al.³ extended the model to the predominantly lateral flows, including the laminar flow conditions, which is important for the modeling of a flow blockage because strong recirculating wakes are induced downstream from the blockage and the flow characteristics change drastically from forced convection to mixed convection.

The distributed resistance model adds the resistance terms in both the axial and the lateral momentum equations; thus, it represents the actual pressure and flow distributions in the subchannels. The forces exerted on the fluid for each direction can be divided into four components, as depicted in Fig. 1. The detailed definitions of the components are described by Ninokata et al.³ and Ha et al.¹⁰ With the expressions of the distributed resistance

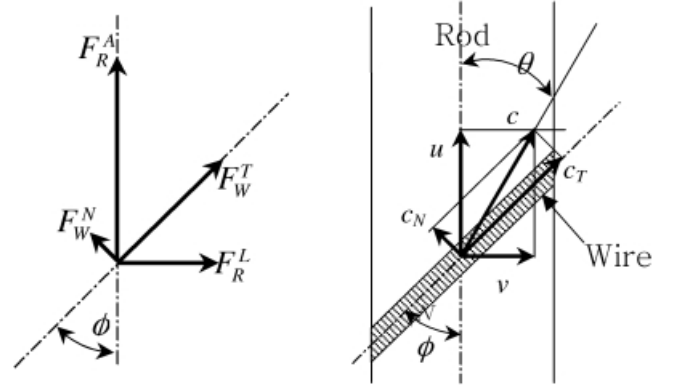


Fig. 1. Components of the drag forces and velocities of a wire-wrapped rod.

of the wire wrap, the last terms of Eqs. (1) and (2) are replaced with the following forms:

$$\begin{aligned} & \frac{1}{2} \left(\frac{f}{D_h} + \frac{K}{\Delta x} \right) \langle \rho u^2 \rangle_A A \\ & = \frac{1}{\Delta x} \left\{ \frac{A_R f}{8} \rho c |c| \cos \theta \right. \\ & \quad \left. + \frac{A_W f}{8} \rho c |c| \cos(\phi - \theta) \cos \phi + F_W^N \sin \phi \right\} \quad (3) \end{aligned}$$

and

$$\begin{aligned} & \frac{1}{2} \frac{S}{l} K_G \langle \rho v^2 \rangle_S \\ & = \frac{1}{l \Delta x} \left\{ F_R^L + \frac{A_W f}{8} \rho c |c| \cos(\phi - \theta) \right. \\ & \quad \left. \times \sin \phi - F_W^N \cos \theta \right\}. \quad (4) \end{aligned}$$

The forces F_R^L and F_W^N are estimated by the correlations depending on the direction of the dominant flow, which is also suggested by Ninokata et al.³ and Ha et al.¹⁰ The expressions of the distributed resistance in Eqs. (3) and (4) are well implemented in the MATRA-LMR-FB code. The effect of the model is evaluated in a later section by a comparison with the original forced cross-flow model.

III.B. Hybrid Difference Scheme

To get the numerical solutions from the thermal-hydraulic governing equations, differential equations like Eqs. (1) and (2) have to be discretized into the difference equations using a proper differencing method. Most

thermal-hydraulic codes adopt the staggered grid method for the definition of the variables and the upwind spatial difference scheme for the differencing of the convective terms. The staggered grid method defines the scalar properties such as pressure, density, and temperature at the center of the control volume and the vector properties such as the velocity at the boundary of the control volume. With the upwind scheme the convective scalar properties are determined depending on the direction of the velocity at the boundary. For example, the enthalpy convected by the cross-flow rate w_j in the energy equation, denoted by h^* , is defined as

$$h_j^* = h_k \quad \text{if } w_j > 0$$

and

$$h_j^* = h_l \quad \text{if } w_j < 0, \quad (5)$$

where k and l are the adjacent channel indices and the positive direction for cross flow is from k to l .

The upwind difference scheme is known to enhance the numerical stability but induces numerical diffusion when there are sharp gradients across the computational mesh. The numerical diffusion varies reciprocally to the convective velocity. That is, it decreases with a higher velocity and increases with a lower velocity. When a blockage exists in a flow path, the flow rate is drastically reduced and recirculating wakes are induced to a certain distance downstream from the blockage. Therefore, the flow fields spread from the Stokes flow to the turbulent flow through the laminar region. If we apply the upwind difference scheme consistently for the whole region of the flow in the subassembly with a blockage, a large numerical diffusion is inevitable in some part of the flow region, especially in the recirculating wake region.

To avoid the unphysical numerical diffusion in the analysis of a flow blockage, the spatial differencing method for the convection terms is modified in the MATRA-LMR-FB code. Instead of the upwind scheme, the hybrid differencing scheme is used based on the Reynolds number (Re) and the Peclet number (Pe). That is, the central difference method is applied when Re is less than 2 or Pe is less than 2, based on the analytical result by Macdougall and Lillington.² With the hybrid difference scheme, the convective terms in the axial momentum, the lateral momentum, and the energy equations are expressed as follows, respectively:

$$\begin{aligned} u \frac{\partial m}{\partial x} &= \frac{1}{\Delta x} [(um^*)_{j+1} - (um^*)_j] \quad \text{if } \text{Re} > 2, \\ &= \frac{1}{\Delta x} \left[\frac{1}{2} (m_{j+1}^* + m_j^*) (u_{j+1} - u_j) \right] \quad \text{if } \text{Re} < 2; \end{aligned} \quad (6)$$

$$\begin{aligned} \bar{u} \frac{\partial w}{\partial y} &= \frac{1}{\Delta y} [(\bar{u}w^*)_k - (\bar{u}w^*)_l] \quad \text{if } \text{Re} > 2, \\ &= \frac{1}{\Delta y} \left[\frac{1}{2} (w_k^* + w_l^*) (\bar{u}_k - \bar{u}_l) \right] \quad \text{if } \text{Re} < 2; \end{aligned} \quad (7)$$

and

$$\begin{aligned} V \frac{\partial(\rho u h)}{\partial x} &= [(mh^*)_{j+1} - (mh^*)_j] \quad \text{if } \text{Pe} > 2, \\ &= \left[\frac{1}{2} (h_{j+1}^* + h_j^*) (m_{j+1} - m_j) \right] \quad \text{if } \text{Pe} < 2. \end{aligned} \quad (8)$$

III.C. Turbulent Mixing Model

To enhance the predictability of the flow and temperature distributions in the subchannels using a subchannel analysis code, it is necessary to model as accurately as possible the intersubchannel mixing phenomenon due to the cross flow between the adjacent subchannels. When a single-phase flow exists in the subchannels, the mixing of the mass, energy, and momentum between the subchannels can be divided into two parts, the mixing by the diversion flow due to the pressure gradient and the mixing by the cross flow mainly due to the turbulence effect. It is believed that the diversion flow is modeled appropriately with the distributed resistance model and the axial pressure drop correlations in the MATRA-LMR-FB code. In consideration of the turbulent mixing, there is some room for the enhancement in the predictability of the code because the code still adopts the old type of turbulent mixing correlations used in the COBRA-IV code even though the turbulent mixing for the two-phase flow has been advanced with the equal-volume exchange turbulent mixing with the void drift (EVVD) model.¹¹

In the MATRA-LMR-FB code, the effects of turbulent mixing are taken into account in the axial momentum equation and the energy conservation equation. The following equation is equivalent to the axial momentum conservation of Eq. (1) for an arbitrary subchannel i :

$$\begin{aligned} \frac{\partial m_i}{\partial t} + \frac{\partial}{\partial x} \left(\frac{m_i^2}{\rho A_i} \right) + \sum_j w_{ij} u^* + \sum_j f_T (w'_{ij} u'_i - w'_{ji} u'_j) \\ = -A \frac{\partial p}{\partial x} - A \rho g - \frac{1}{2} \left(\frac{f}{D_h} + \frac{K}{\Delta x} \right) \langle \rho u^2 \rangle_A A. \end{aligned} \quad (9)$$

Similarly, the energy equation for subchannel i is described as follows:

$$\begin{aligned} A_i \frac{\partial}{\partial t} (\rho_i h_i) + \frac{\partial}{\partial x} (m_i h_i) \\ + \sum_j w_{ij} h^* + \sum_j (w'_{ij} h_i - w'_{ji} h_j) = Q, \end{aligned} \quad (10)$$

where Q denotes the heat input per unit length of the flow channel.

In the above equations, the last terms in the left-hand sides represent the contribution by the turbulent mixing between the subchannel i and its surrounding subchannels. Now, the turbulent mixing model in MATRA-LMR and also in COBRA-IV is only to determine the turbulent mixing flow rate w' and the turbulent momentum factor f_T . The turbulent momentum factor is the same as the turbulent Prandtl number. The turbulent mixing flow rate per unit length from subchannel i to j is defined with the single-phase fluctuating velocity v'_{ij} as follows:

$$w'_{ij} \equiv \rho_i S_{ij} v'_{ij} . \quad (11)$$

Usually, the turbulent mixing flow rate through the interface of the subchannels is defined using the turbulent mixing coefficient β as follows:

$$w'_{ij} = \beta S_{ij} \bar{G}_{ij} , \quad (12)$$

where $\bar{G}$ is the average axial mass flux flowing along the subchannels. Therefore, the turbulent mixing coefficient means the ratio of the effective mean velocity to the axial velocity. This mixing coefficient is essentially the same as the gap Stanton number St_g . The turbulent mixing coefficient is normally determined from the thermal mixing test at single-phase conditions. It is evaluated with the following equation from the diffusive energy flux from channels i to j :

$$\beta = \frac{\varepsilon_H}{l} \frac{\bar{\rho}}{\bar{G}} . \quad (13)$$

In the MATRA-LMR code and COBRA-IV, the turbulent mixing coefficient is determined from the direct input as a constant or is given as a function of Re and the geometry as follows:

$$\begin{aligned} \beta &= a Re^b , \\ &= a Re^b \frac{D_h}{S} , \\ &= a Re^b \frac{D_h}{l} . \end{aligned} \quad (14)$$

The main limitation of this turbulent mixing model is the lack of physical bases in selecting a constant value of the mixing coefficient and the constants for Eq. (14). Further, in the analysis of a flow blockage, a constant mixing coefficient does not reflect the various degrees of the mixing dependent on the flow regime. Therefore, the degree of mixing in the recirculating wake region can be distorted severely. The other drawback of the model is that the principal variables in Eq. (14) are not sufficient to describe the actual turbulent mixing phenomenon. Many

recent studies suggest that the turbulent mixing is a function of several variables. That is,

$$\beta = f(Re, D_h, S, l, d) . \quad (15)$$

The correlations having the form of Eq. (15) are found in the works by Kim and Chung,¹² Sadatomi et al.,¹³ Rehme,¹⁴ and Zhukov et al.¹⁵ Kim and Chung assumed that turbulent mixing is composed of a molecular motion, isotropic turbulent motion, and a flow pulsation and suggested the following correlations:

$$\begin{aligned} St_g &= \frac{2}{\gamma^2} \sqrt{\frac{\alpha}{8}} \frac{D_h/d}{S_{ij}/d} \\ &\times \left[\left(\frac{\gamma^2}{2PrRe^{1-\beta/2}} \sqrt{\frac{8}{\alpha}} + \frac{1}{Pr_t} \right) \frac{S_{ij}/d}{b\delta/d} \right. \\ &\left. + a_x \frac{z_{FP}}{d} Str \right] Re^{-\beta/2} , \end{aligned} \quad (16)$$

where α and β are the constants for the friction factor, expressed as $f = \alpha Re^{-\beta}$, and γ is an empirical constant. By the way, the first part of the correlation is identical to the conduction equation for the conductive heat transfer through the fluid itself, which is separately modeled in a subchannel energy equation. Therefore, the pure turbulent mixing is expressed as a sum of the mixing due to isotropic turbulence and the mixing from a periodic flow pulsation, which are written as follows, respectively:

$$St_{g,IT} = \frac{2}{\gamma^2} \sqrt{\frac{\alpha}{8}} \frac{D_h}{S_{ij}} \frac{1}{Pr_t} \frac{S_{ij}}{b\delta} Re^{-\beta/2} \quad (17)$$

and

$$St_{g,FP} = \frac{2}{\gamma^2} \sqrt{\frac{\alpha}{8}} \frac{D_h}{S_{ij}} a_x \frac{z_{FP}}{d} Str Re^{-\beta/2} . \quad (18)$$

The mixing Stanton number for the total turbulent effect is

$$St_{g,K-C} = St_{g,IT} + St_{g,FP} . \quad (19)$$

The original purpose of Kim and Chung was the derivation of the scale relation applicable to various Prandtl number fluid flows, especially for the application to the thermal-hydraulic analysis of liquid metal coolant reactors. However, in Eqs. (17) and (18) describing the pure turbulence effect, the influence of the Prandtl number is not taken into account and the scale relation is not included. If we compare the magnitude of Eq. (17) with that of Eq. (18), it is found that the flow pulsation is a major cause of the mixing over the isotropic turbulence.

Sadatomi et al.¹³ also divided the turbulent mixing into two components of the turbulent diffusion and the convective transfer due to large-scale periodic eddies. They developed a correlation for turbulent diffusion based

on their own experimental data and for the convective transfer including the data of others. The correlation of Sadatomi et al.'s for the fuel bundle in an array is summarized into the following expression:

$$\beta_{Sada} = 0.00562 \left(\frac{D_h}{S_{ij}} \right) \frac{\text{Re}^{-1/8}}{F_i^*} + 0.0018 \left(\frac{S_{ij}}{d} \right)^{-0.52}, \quad \text{for a square array} \quad (20)$$

$$= 0.00562 \left(\frac{D_h}{S_{ij}} \right) \frac{\text{Re}^{-1/8}}{F_i^*} + 0.000023 \left(\frac{S_{ij}}{d} \right)^{-1.86}, \quad \text{for a triangular array} \quad (21)$$

In the above equations, F_i^* is the subchannel geometry factor explaining the change in the gap clearance along the distance between the centroids of the subchannels. The correlation of Sadatomi et al. for the triangular array results in a very small mixing due to the convective transfer when the ratio of S_{ij}/d is greater than 0.1, which has to be investigated further because the correlation is derived with a very limited number of data in the triangular array bundles.

Through the intensive studies on the structure of the turbulence in the subchannels of rod bundles, Rehme¹⁴ concluded that the natural mixing between the subchannels mainly results from the periodic flow pulsations and that the secondary flow motion does not contribute significantly to the mixing process. He derived the following correlation based on a large number of experimental data and investigation results from the early 1960s to 1990:

$$w'_{ij} = \frac{\rho S_{ij} \bar{\varepsilon}}{\delta_{ij}} Y, \quad Y = \frac{0.7}{(S_{ij}/d)},$$

and

$$\bar{\varepsilon} = \frac{\nu}{20} \text{Re} \sqrt{\frac{\lambda_T}{8}}, \quad \lambda_T = 0.18 \text{Re}^{-0.2}. \quad (22)$$

This correlation can be expressed in the form of the turbulent mixing coefficient through some mathematical manipulations. The following is a different expression of the Rehme correlation:

$$\beta_{Rehme} = 0.00525 \left(\frac{d}{\delta_{ij}} \right) \left(\frac{D_h}{S_{ij}} \right) \text{Re}^{-0.1}. \quad (23)$$

The correlation suggested by Zhukov et al.¹⁵ is of particular importance because it has been based on the experimental data for liquid metal flows in rod bundles. The proposed correlation of a gap Stanton number for the triangular array is as follows:

$$\text{St}_{g,Zh} = \frac{0.393 \left[\left(\frac{2\sqrt{3}}{\pi} \right) \left(\frac{P}{d} \right)^2 - 1 \right]}{\text{Pe}^{0.7} \left(\frac{P}{d} \right) \sqrt{\frac{P}{d} - 1}}. \quad (24)$$

The applicable ranges of the above correlation are $1.10 < P/d < 1.35$, $70 < \text{Pe} < 1600$, and $\text{Pr} \ll 1$.

For the development of the MATRA-LMR-FB code, the above state-of-the-art turbulent mixing models are implemented in the code. To remove the user dependence in selecting the turbulent momentum factor f_T and to complete the turbulent mixing model in MATRA-LMR-FB, a correlation by Kays¹⁶ for the turbulent Prandtl number is used, which is as follows:

$$\text{Pr}_t = \frac{0.7}{\text{Pe}_t} + 0.85, \quad (25)$$

where Pe_t is the turbulent Peclet number, defined as $(\varepsilon_M/\nu)\text{Pr}$.

IV. ANALYSIS OF A FLOW BLOCKAGE

The effects of each model implemented in the MATRA-LMR-FB code for the modeling of a flow blockage are evaluated with the experimental data obtained from the Oak Ridge National Laboratory (ORNL) 19-pin fuel failure mockup (FFM) 5B bundle.¹⁷ The fuel rods, whose diameter is 5.842 mm, are arranged in a triangular pitch within a hexagonal duct. The diameter of the wire wrap around the internal rods is 1.4224 mm and its diameter around the peripheral rods near the hexagonal duct is 0.7112 mm. As shown in Fig. 2, the sodium flow enters from the bottom and passes the entrance region of about 406.4 mm, then the heated section, and finally the exit region where the thermocouples are located for the measurement of the outlet temperature distribution. The length of the heated section is 457.2 mm and the exit region is about 152.4 mm. The thermocouples are located at the middle of the exit region. A plate type of blockage is positioned at 101.6 mm from the start of the heated section. About one-third of the flow area is blocked at the edge around the corner subchannels, 37, 42, and 41 in Fig. 3.

The models are analyzed for the two sets of experimental data, high flow and low flow. In the high-flow case, FFM Series 6, Test 12, Run 101, the inlet temperature was 323.6°C and the inlet velocity was 6.93 m/s. The total heat input from the 19 rods was 145 kW, equivalent to 90.9 W/cm² per rod. For the simulated low-flow case, FFM Series 6, Test 12, Run 109, the inlet sodium temperature was 268.4°C and the inlet velocity was 0.48 m/s. The total amount of heat added to the sodium was 52.8 kW.

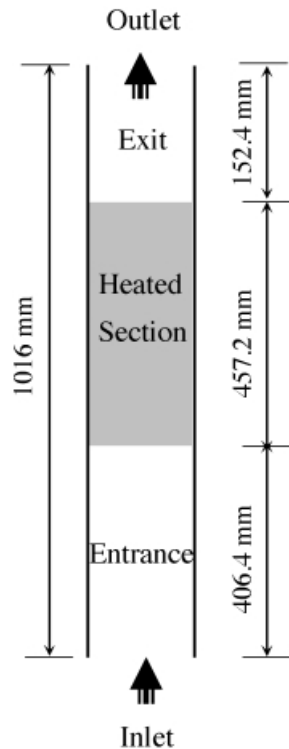


Fig. 2. Schematic diagram of the FFM-5B test section.

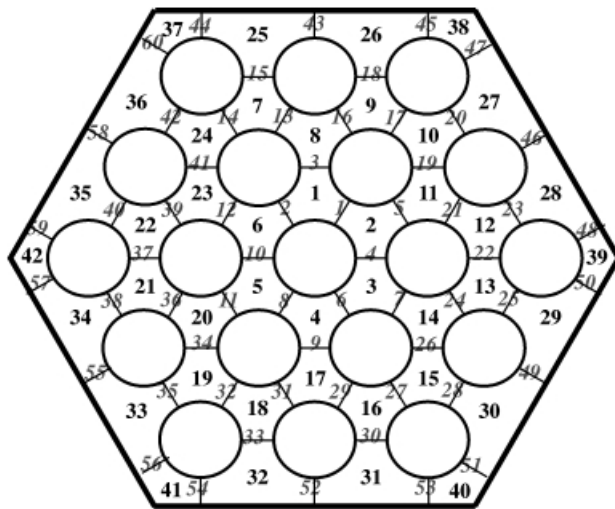


Fig. 3. Subchannel and gap numbering for MATRA-LMR-FB.

IV.A. Effects of the Blockage Models

The effects of the distributed resistance model and the hybrid difference scheme are depicted in Figs. 4 and 5 for the high- and low-flow cases, respectively. The experimental data show clearly that the outlet temperature downstream from the blockage increases when there

is a blockage in the flow path. With a high flow, as shown in Fig. 4, the general trend of the outlet temperature profile is predicted reasonably with the wire forcing function, but the magnitude of the peak temperatures is underestimated quite a lot because the lateral flow rates are estimated in the wrong manner. The distributed resistance model with the upwind scheme gives much improved predictions but still fails to predict the location of the highest temperature, especially at the boundary of the blockage plate. This may be from the inefficiency in the modeling of the recirculating wakes after the blockage. With the distributed resistance model combined with the hybrid difference scheme, it is clearly shown in the figure that the degree of temperature increase due to the blockage agrees well with the data, within 5°C. The overall temperature difference between the blocked region and the unblocked region, which amounts to about 20°C, is calculated well with these models even though the temperatures at the blockage boundary region are slightly overpredicted.

The model predictions for the low-flow case are shown in Fig. 5. As shown in the figure, the result using the wire forcing function agrees well with the experimental data, except for the data on the opposite side of the blockage plate, subchannels 12 and 39. This implies that the assumption of wire forcing is partly valid when the flow rate is rather small. The distributed resistance model also generally follows well all the data profiles. The degree of the temperature discrepancy in the unblocked subchannels, 12 and 39, is corrected with the distributed resistance model. The effect of the hybrid scheme is not so dominant in this case, but it is notable that the temperature distribution at the edge of the blockage and the temperature gradients at the boundary of the blocked and the unblocked subchannels are improved slightly. It can be understood that the wake flow is not so strong when the flow rate is low; thus, the model influences are also not so remarkable with the low inlet flow rate.

All the above model evaluations are performed with a constant value of the turbulent mixing coefficient, 0.01, to analyze the effects of each model independently. However, as described previously the application of a constant mixing coefficient does not reflect the changes of the local mixing depending on the flow regime. Therefore, the state-of-the-art turbulent mixing models are implemented in the MATRA-LMR-FB code and their influences on the analysis of a flow blockage are evaluated.

In Fig. 6, the predicted outlet temperatures with different turbulent mixing models are compared for the high-flow case. The Re at the center subchannels is about 58 000 in this case. The models of Rehme and Zhukov et al. predict the data more accurately than the other models. The models by Kim and Chung and Sadatomi et al. underestimate the mixing downstream from the blockage and thus result in higher outlet temperatures. Kim and Chung's model includes some experimental coefficients whose appropriateness should be investigated

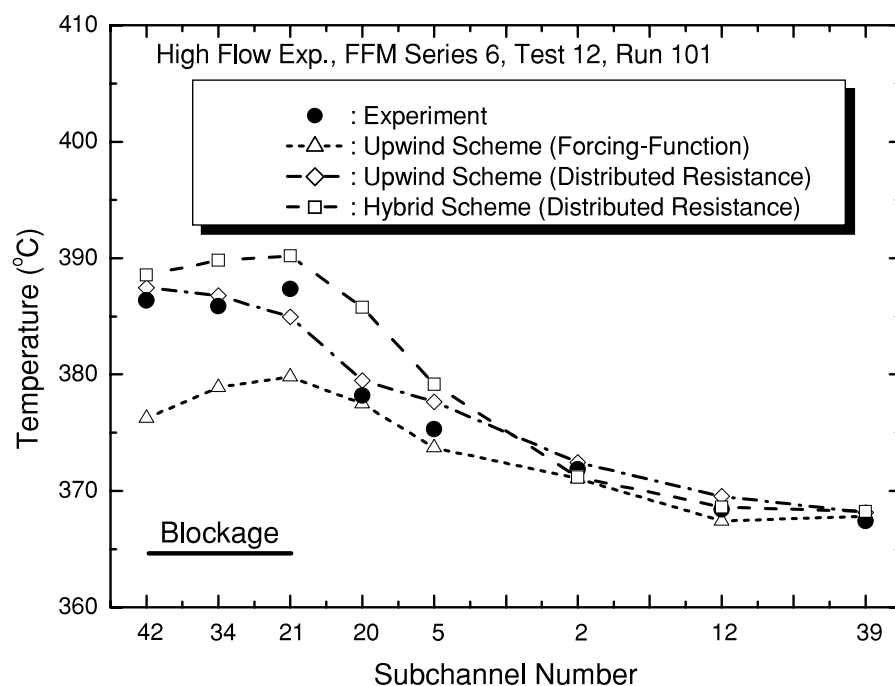


Fig. 4. Effects of the distributed resistance model and hybrid scheme for the high-flow case.

further. On the other hand, the Sadatomi et al. model is analyzed to underestimate the turbulent mixing due to the flow pulsation described in Eq. (21) for the triangular arrayed bundle. To show the improvements by the phe-

nomenological turbulent mixing models in the flow blockage analysis, we compare the predictions with the Rehme model with those with a constant mixing coefficient, 0.01, in Fig. 7. The enhancement in the predictions is seen

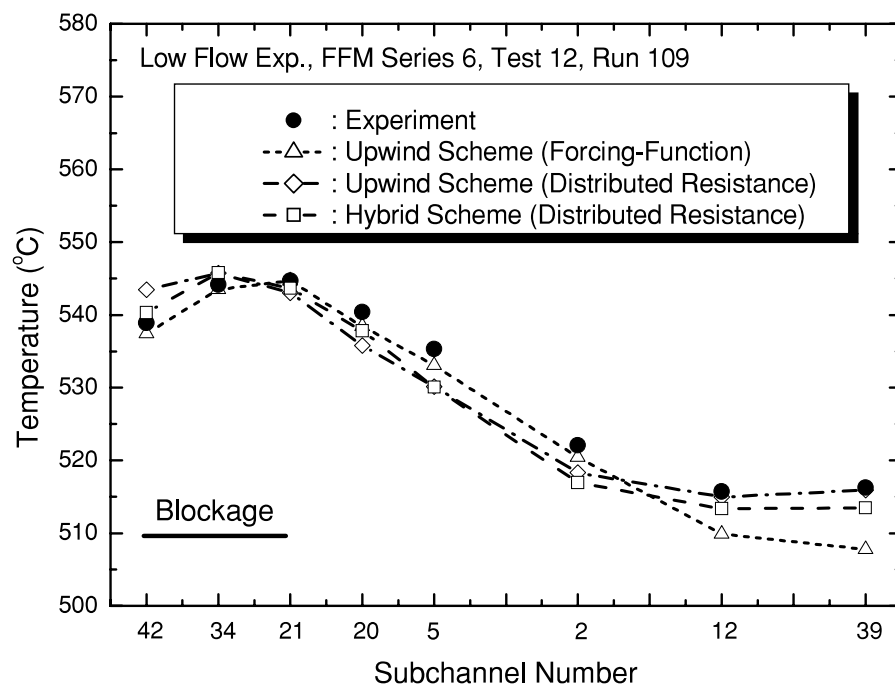


Fig. 5. Effects of the distributed resistance model and hybrid scheme for the low-flow case.

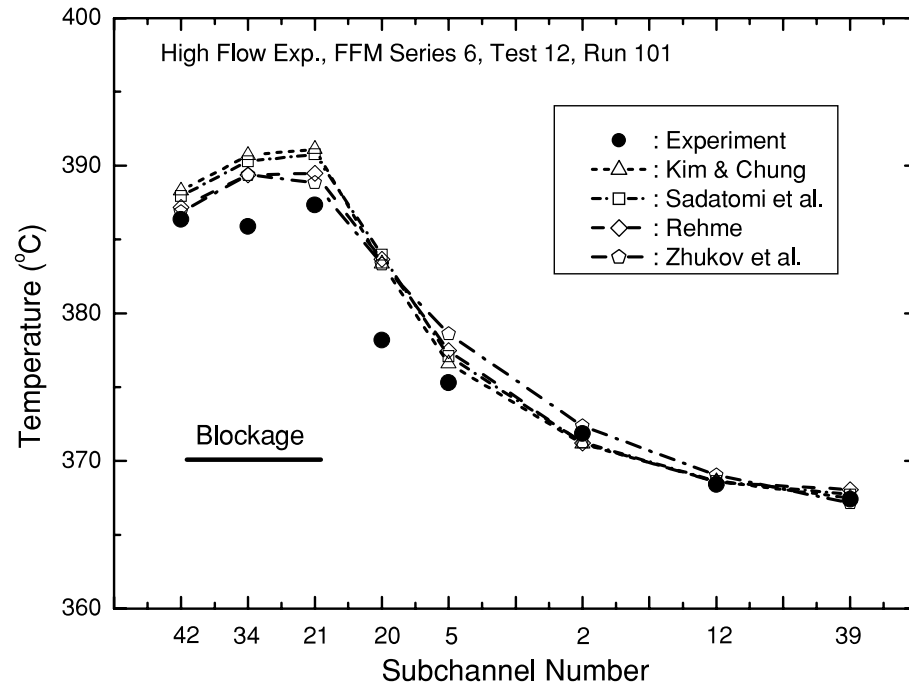


Fig. 6. Effects of the turbulent mixing models for the high-flow case.

clearly at the edge of the blockage and also at the boundary of the blocked and the unblocked subchannels.

The predictions for the low-flow case with the enhanced mixing models are shown in Fig. 8. The Re for

this case is about 3800 and the Pe is about 23. It is found that the Zhukov et al. model is not applicable for the case because the fluidic condition deviates from the suggested ranges in Eq. (24). Figure 8 also shows that the dependence

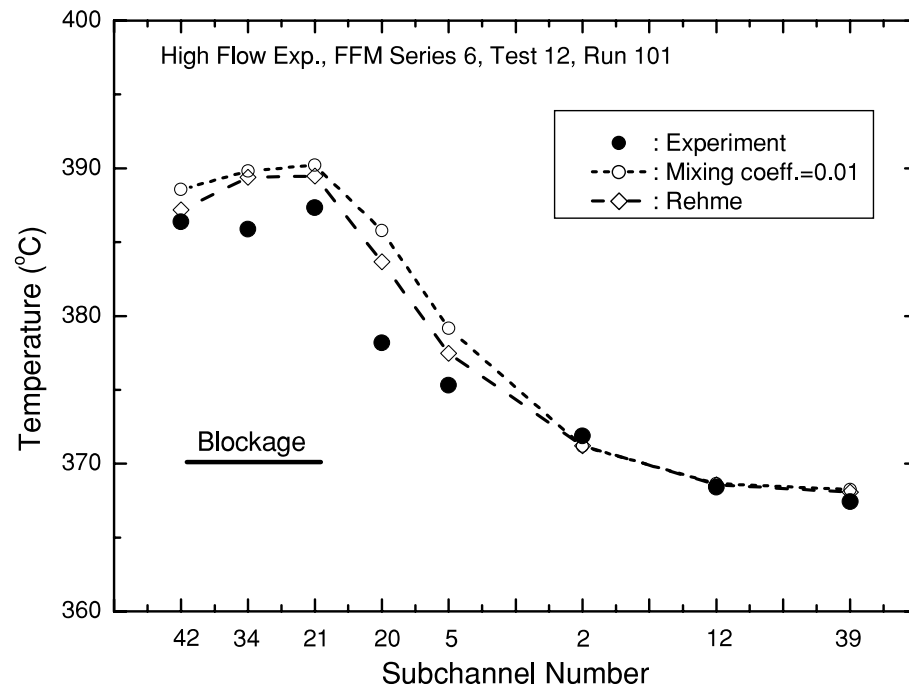


Fig. 7. Predictions with the Rehme model and the constant mixing coefficient for a high flow.

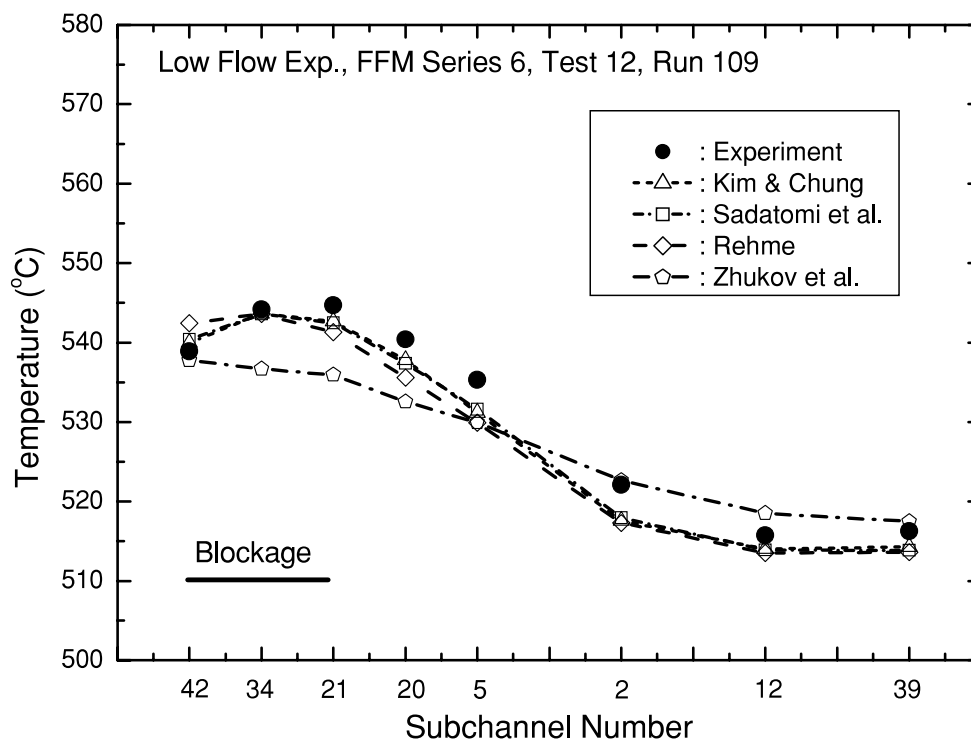


Fig. 8. Effects of the turbulent mixing models for the low-flow case.

on the turbulent mixing model in the prediction of a flow blockage is not dominant at this low-flow condition. The correlations by Kim and Chung and Sadatomi et al. show slightly better results than Rehme's. The predictions with Kim and Chung's model result in nearly the same temperature distributions with a constant mixing coefficient of 0.01, as shown in Fig. 9.

The accuracies of the turbulent mixing models are evaluated indirectly through the above predictions of the experimental data. Therefore, it is difficult to quantitatively compare the predicted mixing coefficients with the models. For a direct comparison of the models with each other and to investigate the appropriateness of each model in the analysis of a flow blockage of an LMR, we computed the turbulent mixing coefficients at the operating condition of the KALIMER with the selected four turbulent mixing models. The evaluated turbulent mixing coefficients with the models for the KALIMER operating condition are summarized in Table I. In the table, it is clearly shown that the correlations by Kim and Chung and Sadatomi et al. predict less turbulent mixing than do those of Rehme and Zhukov et al. Based on the previous evaluation of the ORNL 19-pin data for the high-flow case, the Rehme and Zhukov correlations are more recommendable for the analysis of a blockage in an assembly of an LMR if the inlet velocity is not very low.

IV.B. Comparison with COMMIX Predictions

The porous medium approach is more generally applicable to model the flow in solid objects, such as a flow blockage. It is also expected that the accuracy of modeling with the porous medium method is better than that with the subchannel approach. To validate the flow blockage models in the MATRA-LMR-FB code, we compared the predictions from the code with the results predicted with the COMMIX-1A code, which is one of the well-known porous medium approach codes. The experimental data from the ORNL 19-pin FFM 5B bundle are also used for the comparison. The COMMIX results are obtained from the study by Domanus et al.⁵

The comparisons for the high- and low-flow cases are shown in Figs. 10 and 11, respectively. With high flow, Fig. 10, both COMMIX and MATRA-LMR-FB overestimate the peak temperature downstream from the blocked subchannels. The COMMIX fails to predict the location of the peak temperature while the MATRA-LMR-FB successfully predicts the subchannel with the peak outlet temperature. It is presumed that the modeling of the wire wrap was not sufficient with COMMIX, while the distributed resistance model in MATRA-LMR-FB describes the effect of the wire wrap rather accurately. This superiority in the wire wrap modeling results in a more accurate prediction of the temperature distributions

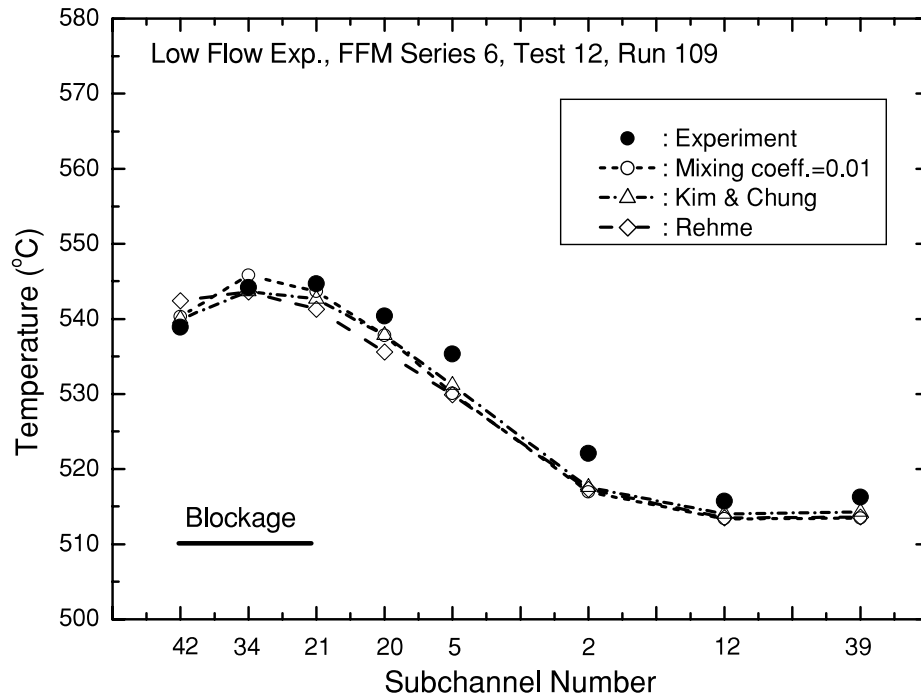


Fig. 9. Predictions with phenomenological models and a constant mixing coefficient for a low flow.

TABLE I
Turbulent Mixing Coefficients for the Operating Condition of KALIMER

Model	Description	Mixing Coefficient
Kim and Chung	$St_{g,K-C} = St_{g,IT} + St_{g,FP}$ $St_{g,IT} = \frac{2}{\gamma^2} \sqrt{\frac{\alpha}{8}} \frac{D_h}{S_{ij}} \frac{1}{Pr_t} \frac{S_{ij}}{b\delta} Re^{-\beta/2}$ $St_{g,FP} = \frac{2}{\gamma^2} \sqrt{\frac{\alpha}{8}} \frac{D_h}{S_{ij}} a_x \frac{z_{FP}}{d} Str Re^{-\beta/2}$	0.002398
Sadatomi et al.	$\beta_{Sada} = 0.00562 \left(\frac{D_h}{S_{ij}} \right) \frac{Re^{-1/8}}{F_i^*} + 0.000023 \left(\frac{S_{ij}}{d} \right)^{-1.86}$	0.003522
Rehme	$\beta_{Rehme} = 0.00525 \left(\frac{d}{\delta_{ij}} \right) \left(\frac{D_h}{S_{ij}} \right) Re^{-0.1}$	0.007071
Zhukov et al.	$St_{g,Zh} = \frac{0.393 \left[\left(\frac{2\sqrt{3}}{\pi} \right) \left(\frac{P}{d} \right)^2 - 1 \right]}{Pe^{0.7} \left(\frac{P}{d} \right) \sqrt{\frac{P}{d} - 1}}$ (1.10 < P/d < 1.35, 70 < Pe < 1600, Pr ≪ 1)	0.006320

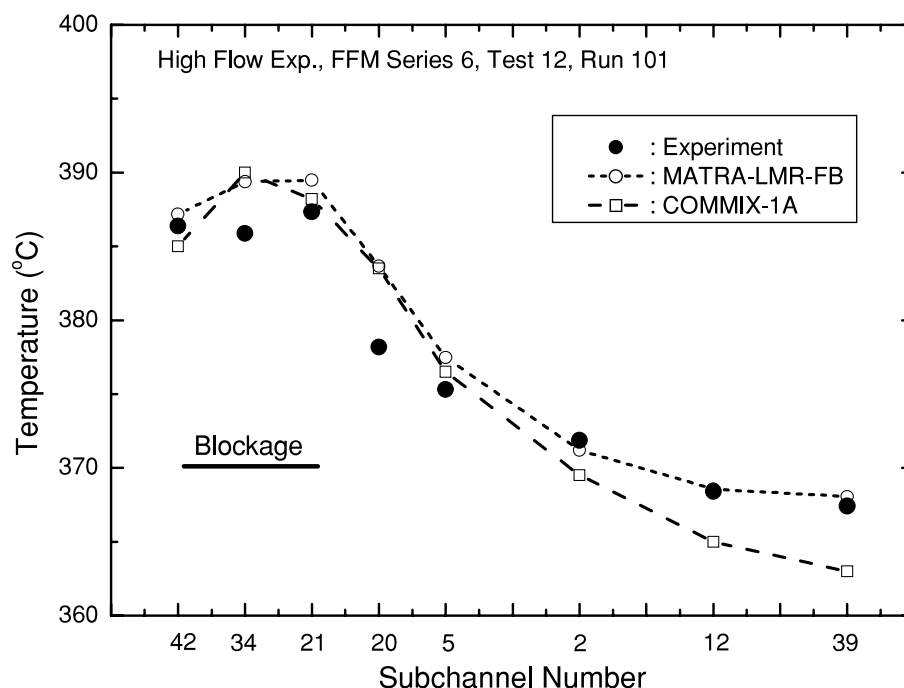


Fig. 10. Comparison of the MATRA-LMR-FB prediction with COMMIX-1A for a high flow.

in the unblocked subchannels, 2, 12, and 39. In Fig. 11, it is found that the prediction for the low flow with the COMMIX agrees well with the experimental data, except for the outermost subchannels, 42 and 39. The

MATRA-LMR-FB code also predicts the general trend with a reasonable accuracy, but the temperatures at the boundaries between the blocked and the unblocked subchannels are slightly underestimated. It is again confirmed

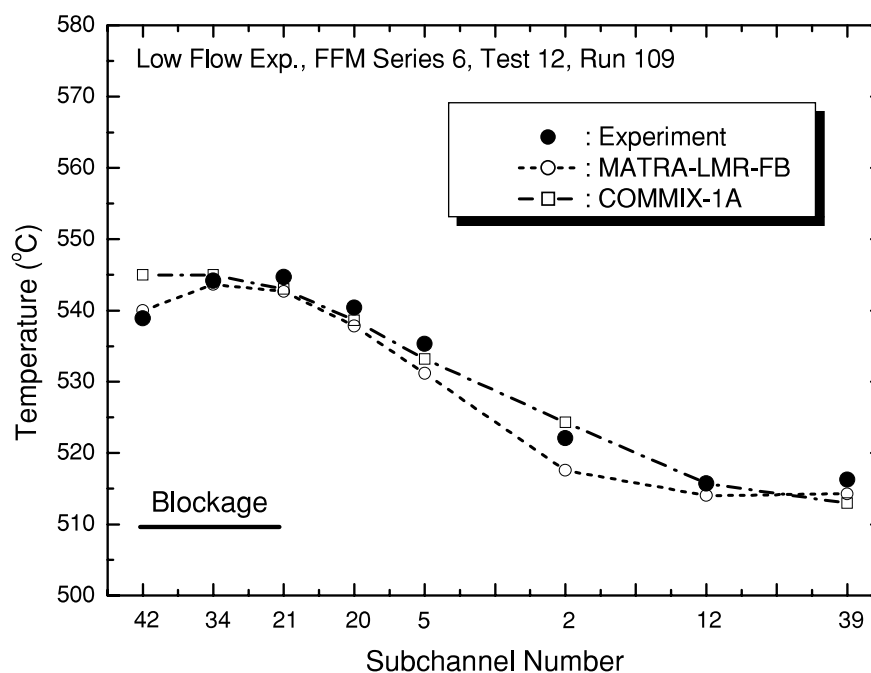


Fig. 11. Comparison of the MATRA-LMR-FB prediction with COMMIX-1A for a low flow.

that the sweeping flow along the outer subchannels adjacent to the hexagonal duct, which is a typical flow characteristic in the flow path with a wire wrap, is well represented with the distributed resistance model in MATRA-LMR-FB. Because the direction of the lateral velocities is ignored in the subchannel approaches, it seems difficult to predict the temperatures in the boundary subchannels between the blocked and the unblocked regions accurately when the flow is low. Even though the accuracy of MATRA-LMR-FB in a low-flow case is slightly lower than that of COMMIX, the predictability of the MATRA-LMR-FB code for a flow blockage is generally evaluated to be comparable to that of the COMMIX code.

IV.C. Blockage Analysis for the KALIMER Design

The safety of the KALIMER⁶ fuel assembly against a flow blockage is analyzed with the MATRA-LMR-FB code, and the results are compared with the predictions by the SABRE4 code. The preliminary design of the KALIMER, a 150-MW(electric) pool-type sodium-cooled prototype reactor, has been completed by the Korea Atomic Energy Research Institute (KAERI). The KALIMER core system is designed to generate 392.2 MW(thermal) of power. The reference core utilizes a homogeneous core configuration in the radial direction with two driver fuel enrichment zones, surrounded by a layer of blanket assemblies. The reference core has an active core height of 100 cm and a radial equivalent diameter of 172 cm, and the height-to-diameter ratio for the active core is 0.581. The physically outermost core diameter of all the assemblies is 344.7 cm. The major design parameters and geometric characteristics of the assembly are given in Table II. A KALIMER driver fuel assembly was chosen to calculate the temperature distribution with the MATRA-LMR-FB and to compare the results with the SABRE4 code. The configuration of a KALIMER subassembly and the subchannel numbers used in the calculations are depicted in Fig. 12.

The effects of blockages on reactor safety depend on several factors such as the size and thermophysical properties of the blockage, location of the blockage in the assembly, fuel pin power, and coolant velocity in the assembly. In the present analysis, the area and radial location of the blockage are taken into account. To evaluate the effect of a blockage area, we analyzed 6-channel, 24-channel, and 54-channel blockages. The locations of a blockage considered are the center, the midpoint, and the edge. The three different locations of the 6-channel blockage are shown in Fig. 12. Nine calculations are performed at a uniform pin power of 227.1 W/cm with the mean coolant velocity of about 5.5 m/s, which represents the design operation condition of the driver fuel assembly of KALIMER.

Figure 13 shows the typical sodium temperature profiles just above the blockage and at the exit for the

TABLE II
Design Parameters for the KALIMER Core

Component	Parameter	Design Value
Core	Core thermal output [MW(thermal)]	392.2
	Core electric power [MW(electric)]	150.0
	Core inlet/outlet temperatures (°C)	386.2/530.0
	Total flow rate (kg/s)	2143.1
	Active core height (mm)	1000.0
	Core diameter (mm)	3460.8
	Core configuration	Homogeneous
	Pins per assembly (driver)	271
	Total axial height (mm)	3163.0
	Rod outer diameter (mm)	7.4
Driver fuel rod	Rod pitch (mm)	8.9
	Wire wrap diameter (mm)	1.4
	Wire wrap lead (mm)	204.9
	Cladding thickness (mm)	0.55
	Duct wall thickness (mm)	3.7
	Duct inside flat-to-flat distance (mm)	149.6
	Nominal linear pin power (W/cm)	227.1
	Assembly nominal flow rate (kg/s)	27.2
	Assembly coolant inlet temperature (°C)	386.2
	Number of axial nodes	103
	Radial power distribution	Uniform

6-channel center blockage. The temperature predictions along the axial level are very similar; however, the results of SABRE4 are slightly higher than those of MATRA-LMR-FB, especially at the peripheral subchannels. These trends are observed in all the simulations, except for the calculation of the 54-channel blockage. As shown in Fig. 14, the two code results are in good agreement in the 6- and 24-channel blockages. However, for the 54-channel blockage, the differences between the two codes are rather large. The SABRE4 code computes the peak temperature of 824°C and the MATRA-LMR-FB code calculates the temperature of 671°C for the blocked fraction of 0.09, i.e., for a 54-channel blockage. It is presumed that the SABRE4 code overestimates the residence time of the sodium coolant in the recirculation region downstream from the blockage for the 54-channel blockage. The sodium temperatures predicted by the two codes are all much lower than the boiling temperature; thus, the safety of the KALIMER design is guaranteed for the flow blockage.

The experimental investigation of the German sodium-cooled fast reactor (SNR) for the blocked fraction of 0.147 provides a good reference for the above calculation for the 54-subchannel blockage of the KALIMER. The peak temperature obtained in the SNR was about 780°C (Ref. 18). The inlet temperature and the mass flow rate of the SNR blockage test were 380°C and 23 kg/s, respectively, which are very similar to the specifications of the KALIMER design. It is a general trend that a higher peak temperature is obtained with a higher blockage fraction. Therefore, the peak temperature of 671°C predicted with the MATRA-LMR-FB for the

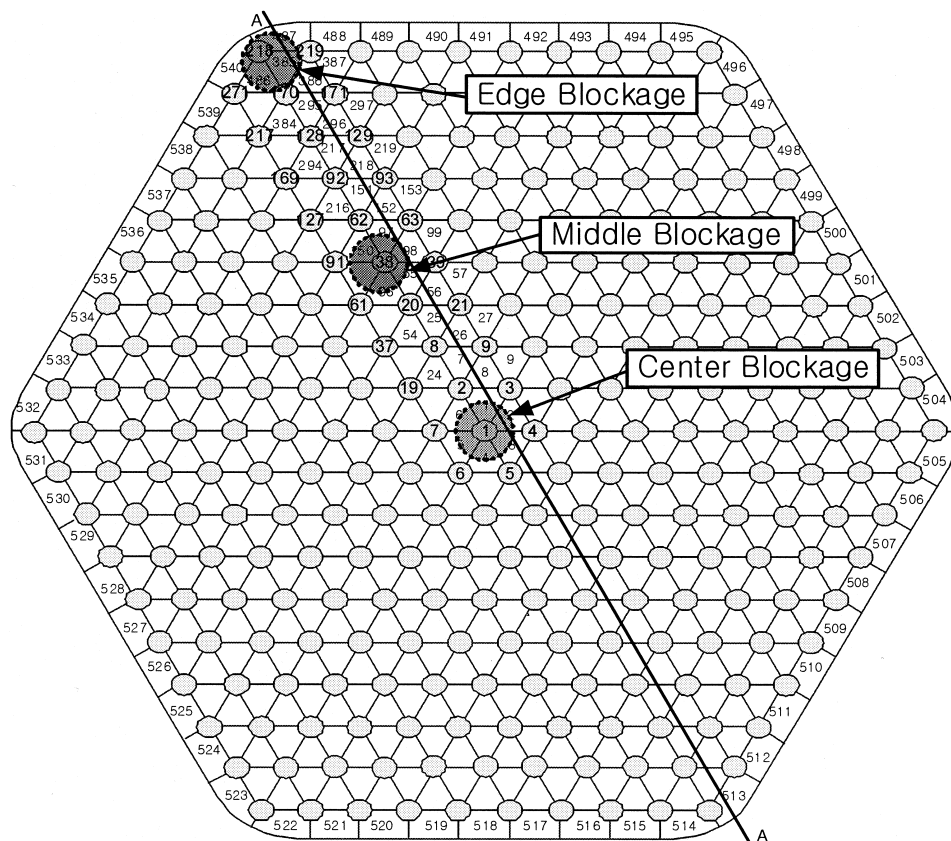


Fig. 12. Subchannel numbering of the MATRA-LMR-FB for the KALIMER 271-pin fuel assembly.

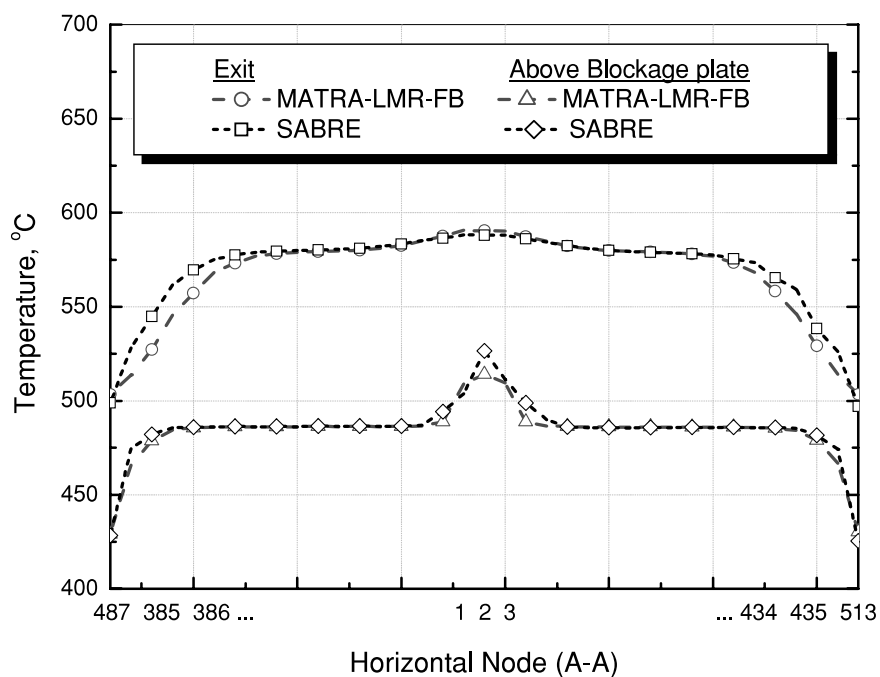


Fig. 13. Temperature profiles for the KALIMER subassembly with a 6-subchannel center blockage.

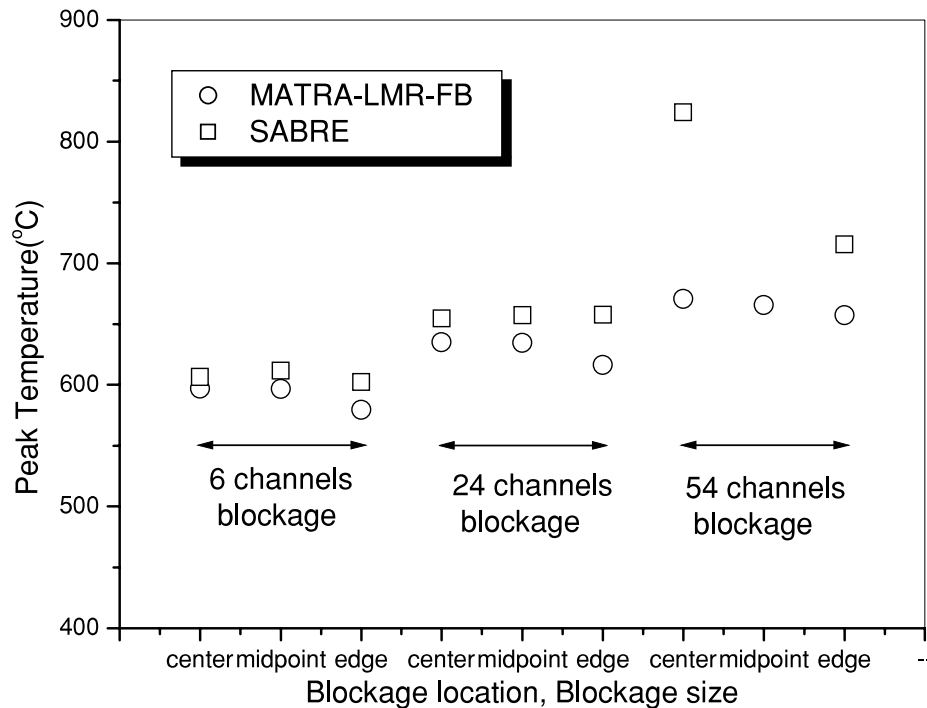


Fig. 14. Peak temperatures predicted for various blockage sizes and locations.

blocked fraction of 0.09 is consistent with the SNR data of 780°C for the blocked fraction of 0.147. It also should be noted that the mass flow rate for a KALIMER subassembly is higher than that of the SNR. It might be concluded that the MATRA-LMR-FB gives more precise predictions than the SABRE4 for the cases.

V. CONCLUSIONS

The model requirements for the analysis of a flow blockage with a subchannel analysis code are studied. The required models are the distributed resistance model to take into account the impact of the wire wrap, the hybrid spatial difference scheme to minimize the numerical diffusion in the wake region downstream from the blockage, and the turbulent mixing model for the exchanges of momentum and energy at the boundaries of the subchannels.

The models are implemented in the MATRA-LMR-FB code and the effects of each model on the prediction of the ORNL 19-pin experiments are investigated step by step. The distributed resistance model and the hybrid difference scheme enhance considerably the prediction of the peak temperature and temperature distribution at the outermost subchannels. The effects of the state-of-the-art turbulent mixing models are also analyzed and the Rehme and Zukov et al. models are found to be most appropriate for the description of a flow block-

age in an LMR subassembly if the flow rate is not very low. By adopting these mixing models, it becomes possible to remove the user dependence in the selection of the turbulent mixing parameters in a subchannel approach code.

The predictability of the MATRA-LMR-FB code with the improved blockage models is comparable to the capability of the COMMIX-1A. Further, the peak temperatures and the temperature distributions calculated for the KALIMER subassembly with various blockage sizes and locations by the MATRA-LMR-FB are consistent with those predicted by the SABRE4 code for the 6- and 24-channel blockages. The MATRA-LMR-FB gives more reasonable results than SABRE4 in the evaluation of the 54-channel blockages. Therefore, it is concluded that the modeling efforts for the analysis of a flow blockage are successfully integrated in the development of the MATRA-LMR-FB code. Further research and development will be concentrated on the modeling of the total inlet blockage of a subassembly in an LMR with the code.

NOMENCLATURE

- A = subchannel flow area
- A_R = rod surface area within the control volume
- A_w = wire wrap surface area within the control volume
- a_x = directional velocity scale of the flow pulsation

b	= shape factor for the length scale for the transverse flow pulsation	w	= lateral flow rate
C_S	= factor accounting for the imperfect coupling between communicating gaps	w'	= fluctuating lateral flow rate
c	= flow velocity	w_{ij}	= lateral flow rate from subchannel i to j
$[D_C]$	= matrix for interchannel connection	x	= coordinate of axial direction
D_h	= hydraulic diameter of a subchannel	y	= coordinate of lateral direction
d	= rod diameter	z_{FP}	= hypothetical path length of flow pulsation
F_i^*	= subchannel geometry factor	<i>Greek</i>	
F_R^A	= axial component of the force exerted by the rod surface	α	= constant for friction factor
F_R^L	= lateral component of the force exerted by the rod surface	β	= turbulent mixing coefficient, constant for friction factor
F_W^T	= tangential component of the force exerted by the wire wrap surface	δ	= distance between the centers of two adjacent subchannels
F_W^N	= normal component of the force exerted by the wire wrap surface	Δ	= node size
f	= friction factor in rod bundle without wire wrap	ε_H	= eddy diffusivity for energy
f_T	= turbulent momentum factor	ε_M	= eddy diffusivity for momentum
G	= axial mass flux	$\bar{\varepsilon}$	= reference eddy viscosity
g	= gravitational constant	ϕ	= angle between the wire wrap and fuel rod
h	= enthalpy	γ	= empirical constant in Eq. (16)
K	= form loss coefficient	ν	= kinematic viscosity
K_G	= lateral loss coefficient	θ	= angle between the flow and the fuel rod
l	= distance between the center of the two adjacent subchannels	ρ	= density
m	= axial flow rate	ω	= wire wrap position
P	= rod pitch	<i>Subscripts</i>	
Pe	= Peclet number	A	= area averaged
Pr	= Prandtl number	FP	= flow pulsation
p	= pressure	g	= gap
S	= gap size	i	= subchannel i
St_g	= gap Stanton number	ij	= from subchannel i to j ; between subchannels i and j
Str	= Strouhal's number	IT	= isotropic turbulence
t	= time	j	= subchannel j
u	= axial velocity	$K-C$	= Kim and Chung
u'	= fluctuating component of axial velocity	S	= gap averaged
V	= volume of a control volume	$Sada$	= Sadatomi et al.
v	= transverse velocity	t	= turbulent
v'	= fluctuating transverse velocity	Zh	= Zhukov
		<i>Superscripts</i>	
		T	= transpose

ACKNOWLEDGMENTS

This work was performed under the long-term nuclear R&D programs sponsored by the Korea Ministry of Science and Technology. The authors express many thanks to H. Ninokata at Tokyo Institute of Technology for the helpful discussion on the development of the distributed resistance model.

REFERENCES

1. C. W. STEWART, C. L. WHEELER, R. J. CENA, C. A. McMONAGLE, J. M. CUTA, and D. S. TRENT, "COBRA-IV: The Model and the Method," BNWL-2214, Pacific Northwest Laboratories (1977).
2. J. D. MACDOUGALL and J. N. LILLINGTON, "The SABRE Code for Fuel Rod Cluster Thermohydraulics," *Nucl. Eng. Des.*, **82**, 171 (1984).
3. H. NINOKATA, A. EFTHIMIADIS, and N. E. TODREAS, "Distributed Resistance Modeling of Wire-Wrapped Rod Bundles," *Nucl. Eng. Des.*, **104**, 93 (1987).
4. E. I. H. LIN and W. T. SHA, "Thermal-Hydraulic Analysis of a Wire-Wrapped 19-Rod Bundle with Edge Blockage," *Trans. Am. Nucl. Soc.*, **28**, 539 (1978).
5. H. M. DOMANUS, V. L. SHAH, and W. T. SHA, "Applications of the COMMIX Code Using the Porous Medium Formulation," *Nucl. Eng. Des.*, **62**, 81 (1980).
6. D. HAHN et al., "KALIMER Conceptual Design Report," KAERI/TR-2204/2002, Korea Atomic Energy Research Institute (2002).
7. W. S. KIM, Y. G. KIM, and Y. J. KIM, "A Subchannel Analysis Code MATRA-LMR for Wire-Wrapped LMR Subassembly," *Ann. Nucl. Energy*, **29**, 303 (2002).
8. Y. J. YOO, D. H. HWANG, and D. H. SOHN, "Development of a Subchannel Analysis Code MATRA Applicable to PWRs and ALWRs," *J. Korean Nucl. Soc.*, **31**, 314 (1999).
9. A. L. DAVIS, "SABRE I: A Computer Program for the Calculation of Three-Dimensional Flows in Rod Clusters," AEEW-R 1057, Atomic Energy Establishment, Winfrith (1979).
10. K. S. HA, H. Y. JEONG, W. P. CHANG, Y. M. KWON, and Y. B. LEE, "Wire-Wrap Models for Subchannel Blockage Analysis," *J. Korean Nucl. Soc.*, **36**, 165 (2004).
11. D. H. HWANG, Y. J. YOO, W. K. IN, and S. Q. ZEE, "Assessment of the Interchannel Mixing Model with a Subchannel Analysis Code for BWR and PWR Conditions," *Nucl. Eng. Des.*, **199**, 257 (2000).
12. S. KIM and B. J. CHUNG, "A Scale Analysis of the Mixing Rate for Various Prandtl Number Flow Fields in Rod Bundles," *Nucl. Eng. Des.*, **205**, 281 (2001).
13. M. SADATOMI, A. KAWAHARA, and Y. SATO, "Prediction of the Single-Phase Turbulent Mixing Rate between Two Parallel Subchannels Using a Subchannel Geometry Factor," *Nucl. Eng. Des.*, **162**, 245 (1996).
14. K. RHEME, "The Structure of Turbulence in Rod Bundles and the Implications on Natural Mixing Between the Subchannels," *Int. J. Heat Mass Transfer*, **35**, 567 (1992).
15. A. V. ZHUKOV, P. L. KIRILOV, A. P. SOROKIN, and N. M. MATJUKHIN, "Transverse Turbulent Momentum and Energy Exchange in the Channels of Complicated Form," *Proc. Heat Transfer*, Brighton, United Kingdom, August 14–18, 1994, Vol. 4, p. 327 (1994).
16. W. M. KAYS, "Turbulent Prandtl Number—Where Are We?," *J. Heat Transfer*, **116**, 284 (1994).
17. H. M. DOMANUS and W. T. SHA, "Numerical Results for a Hexagonal Fuel Assembly with a Planar Blockage Using the COMMIX-1A Computer Code," NUREG/CR-0483, U.S. Nuclear Regulatory Commission (1978).
18. D. KIRSCH, "Investigations on the Flow and Temperature Distribution Downstream of Local Coolant Blockages in Rod Bundle Subassemblies," *Nucl. Eng. Des.*, **31**, 266 (1974).